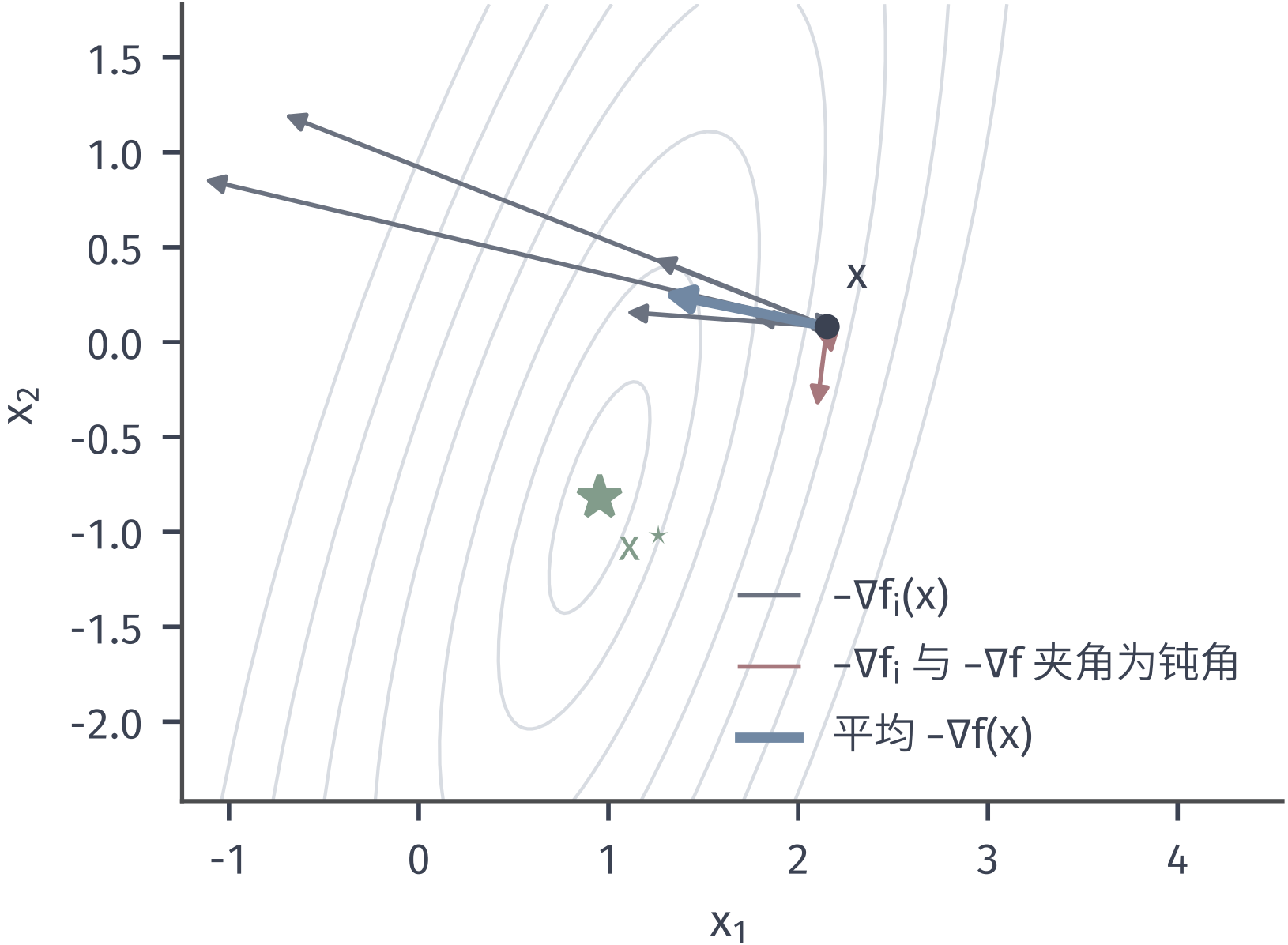
1.5
1.0
0.5
0.0
-0.5
-1.0
-1.5
-2.0
x_2
-1
0
1
2
3
4
x_1
x
$x^\star$
$-\nabla f_i(x)$
$-\nabla f_i$ 与 $-\nabla f$ 夹角为钝角
平均 $-\nabla f(x)$